Week 1
In-Video Quiz Questions

This document contains all of this week's in-video quiz questions. We are providing the questions in this format because we know that some of you would prefer to have a permanent record of the quiz questions to aid your study. Also, if you are downloading the lectures to watch off-line, you do not have access to the in-video quiz questions. Answers to the questions are provided at the end of the quiz. If you miss a question, we encourage you to consult the lecture outline and watch the video again so that the correct answer makes sense to you.

1.1 Basic Heart Anatomy

1. The heart is located in the _____ where it is surrounded by a protective membrane sac called the _____.
   A. abdominal cavity . . . pleurae
   B. thoracic cavity . . . pleurae
   C. mediastinum . . . pericardium
   D. mediastinum . . . pleurae

2. All of the chambers of the heart have muscular walls. The muscle layer is thickest in the _____.
   A. right atrium
   B. right ventricle
   C. left atrium
   D. left ventricle

3. Oxygenated blood returns to the left atrium via the _____. The left ventricle then ejects this oxygenated blood into the _____.
   A. superior vena cava and inferior vena cava . . . pulmonary trunk
   B. pulmonary veins . . . pulmonary trunk
   C. pulmonary veins . . aorta
   D. superior vena cava and coronary sinus . . . aorta

4. The right ventricular wall is thicker than the left ventricular wall because the right ventricle pumps blood through the aorta.
   A. True
   B. False

5. The _____ carries the blood that supplied the walls of the heart back to the right atrium.
   A. superior vena cava
   B. inferior vena cava
   C. coronary sinus
   D. aorta

6. The pulmonary trunk carries deoxygenated blood toward the lungs.
   A. True
   B. False

Answers: 1(C), 2(D), 3(C), 4(B), 5(C), 6(A)

1.2a Your Beating Heart

1. The electrical impulse that generates rhythmic and coordinated contraction of the heart is normally initiated in the _____.
   A. SA node
   B. AV node
   C. AV bundle
   D. right and left bundle branches

2. The short delay in impulse conduction at the _____ allows time for the atria to contract before the ventricles begin contracting.
   A. SA node
   B. AV node
   C. AV bundle
   D. Purkinje fibers

3. The part of the cardiac conduction system that excites contraction of the interventricular septum is the _____.
   A. SA node
   B. AV bundle
   C. right and left bundle branches
   D. Purkinje fibers

4. The only electrical connection between the atria and ventricles is the _____.
   A. SA node
   B. AV node
   C. AV bundle
   D. right and left bundle branches

5. All myocardial cells (i.e., cardiac muscle cells) can generate an electrical signal.
   A. True
   B. False

Answers: 1(A), 2(B), 3(C), 4(C), 5(A)

1.2b ECG Demonstration

1. On an ECG tracing, isoelectric lines represent time periods when there is no change in the electrical activity of the heart.
   A. True
   B. False

2. The ECG waveform that represents ventricular depolarization is the _____.
   A. P wave
   B. QRS complex
   C. T wave

3. The ECG waveform that represents ventricular repolarization is the _____.
   A. P wave
   B. QRS complex
   C. T wave

4. The ECG waveform that represents atrial depolarization is the _____.
   A. P wave
   B. QRS complex
   C. T wave

5. When does the atrium repolarize?
   A. before ventricular depolarization
   B. after ventricular depolarization
   C. during ventricular depolarization
   D. The atrium never repolarizes.

Answers: 1(A), 2(B), 3(C), 4(A), 5(C)

1.3a Heart Valves, Part 1

1. When pressure in the ventricles becomes higher than pressure in the atria, the AV valves _____.
   A. open
   B. close

2. The left AV valve is called the _____.
   A. aortic valve
   B. pulmonic valve
   C. mitral valve
   D. tricuspid valve

3. The valve which prevents blood from flowing backward from the right ventricle to the right atrium during ventricular contraction is the _____.
   A. aortic valve
   B. pulmonic valve
   C. mitral valve
   D. tricuspid valve

4. What thready structures made of tough collagen help to hold the AV valves in the closed position during ventricular contraction?
   A. chordae tendineae
   B. right and left bundle branches
   C. AV bundle
   D. papillary muscles

Answers: 1(B), 2(C), 3(D), 4(A)

1.3b Heart Valves, Part 2

1. Which valve prevents the backflow of blood into the left ventricle during ventricular relaxation?
   A. aortic valve
   B. pulmonic valve
   C. mitral valve
   D. tricuspid valve

2. During the ventricular contraction phase, the first change in valve position is _____.
   A. AV valves close
   B. AV valves open
   C. semilunar valves close
   D. semilunar valves open

3. Closing of the semilunar valves creates the _____ heart sound.
   A. first
   B. second

4. When pressure in the atrium becomes higher than pressure in the ventricle, _____.
   A. the semilunar valve opens
   B. thesemilunar valve closes
   C. the AV valve opens
   D. the AV valve closes

5. Which valve prevents the backflow of blood into the left atrium during ventricular contraction?
   A. aortic valve
   B. pulmonic valve
   C. mitral valve
   D. tricuspid valve

Answers: 1(A), 2(A), 3(B), 4(C), 5(C)

1.3c Assessing Heart Rate Demonstration

1. The act of listening to sounds that arise within body organs, usually with a stethoscope, is called _____.
   A. palpation
   B. auscultation

2. The most accurate way to assess heart rate is to _____.
   A. auscultate at the point of maximal impulse
   B. palpate at the point of maximal impulse
   C. palpate a large artery

3. A heart murmur, which creates a swishing sound, occurs when a heart valve does not close properly.
   A. True
   B. False

4. The point of maximal impulse is where the apex rests between the 5th and 6th rib.
   A. True
   B. False

5. S1 occurs when the _____.
   A. AV valves close
   B. AV valves open
   C. semilunar valves open
   D. semilunar valves close

Answers: 1(B), 2(A), 3(A), 4(A), 5(A)

1.4a Cardiac Cycle, Part 1

1. Which of the following terms is used to denote the series of of events that occur in the heart, each and every time the heart beats?
   A. depolarization
   B. repolarization
   C. the cardiac cycle
   D. electrocardiogram (ECG)

2. Which of the following cells insures that the heart contracts in a rhythmic sequence?
   A. myocardial conducting cells
   B. myocardial contracting cells

3. Isoelectric lines on an ECG tracing represent periods when _____.
   A. all of the myocardial cells are simultaneously depolarizing
   B. all of the myocardial cells are simultaneously repolarizing
   C. there is no change in the electrical state of the myocardial cells

Answers: 1(C), 2(A), 3(C)

1.4b Cardiac Cycle, Part 2

1. During the cardiac cycle, the ventricles hold their largest volume of blood, called _____ at the end of _____.
   A. ESV (end systolic volume) . . . ventricular systole
   B. EDV (end diastolic volume). . . . ventricular diastole

2. The onset of ventricular diastole is associated with the closing of the _____ valves, and the onset of ventricular systole is associated with the closing of the _____ valves.
   A. semilunar. . . .atrioventricular
   B. atrioventricular. . . . semilunar

3. The phase of the cardiac cycle in which the myocardium contracts and ejects blood into the larger arteries (i.e., the aorta and pulmonary trunk) is called _____.
   A. atrial diastole
   B. atrial systole
   C. ventricular diastole
   D. ventricular systole

4. Ventricular systole creates less pressure than atrial systole.
   A. True
   B. False

5. What ECG event represents the depolarization which triggers ventricular systole?
   A. P wave
   B. QRS complex
   C. T wave

Answers: 1(B), 2(A), 3(D), 4(B), 5(B)

1.4c Assessing Pulse Demonstration

1. In people with healthy cardiovascular function, pulse rate is equal to heart rate.
   A. True
   B. False

2. Blood is ejected into the aorta during ventricular systole, creating a wave of pressure in the systemic circulation. This wave creates distension of the blood vessel walls which is called the pulse.
   A. True
   B. False

3. The pulse may be especially strong in a person whose heart is beating very weakly.
   A. True
   B. False

4. Where is the radial artery pulse palpated?
   A. bend of the hip (i.e., the groin area)
   B. wrist
   C. behind the knee
   D. on the foot

Answers: 1(A), 2(A), 3(B), 4(B)

1.5 Cardiac Output

1. Use the following information to answer this question. Heart rate = 70 beats per minute (bpm). End diastolic volume (EDV) = 120 mL. End systolic volume = 50 mL. What is the cardiac output?
   A. 3.5 L/min
   B. 4.9 L/min
   C. 8.4 L/min
   D. 7 L/min

2. What parts of the heart are innervated by the sympathetic nervous system?
   A. AV node, SA node
   B. AV valves, SL valves
   C. SA node, AV valves, AV node
   D. SA node, AV node, myocardial contractile cells

3. What parts of the heart are innervated by the parasympathetic nervous system?
   A. AV node, SA node
   B. SA node, AV valves, AV node
   C. AV valves, SL valves
   D. SA node, AV node, myocardial contractile cells

4. The parasympathetic division of the autonomic nervous system slows the heart rate by _____.
   A. slowing the rate of blood flow back to the heart during diastole
   B. causing the myocardial contractile cells to contract for a longer time
   C. slowing the self-excitation rate of the SA node and prolonging impulse delay in the AV node
   D. causing the myocardial contractile cells to contract more forcefully

Answers: 1(B), 2(D), 3(A), 4(C)